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United States Patent Application Publication	20250279839
Kind Code	A1
Publication Date	September 04, 2025
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ENHANCED UE MAXIMUM SENSITIVITY DEGRADATION INDICATION

Abstract

Method and apparatus for an enhanced user equipment (UE) maximum sensitivity degradation (MSD) indication are provided. The apparatus receives a request for a measurement of an MSD. The apparatus measures the MSD caused by self-interference or a different interference than the self-interference. The apparatus further reports a report including the measurement of the MSD. The measurement of the MSD includes relative information associated with a first amount of the MSD caused by the self-interference.

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Family ID: **95064536**

Appl. No.: **18/892278**

Filed: **September 20, 2024**

Related U.S. Application Data

parent US continuation-in-part 18592389 20240229 PENDING child US 18892278

Publication Classification

Int. Cl.: **H04B17/345** (20150101); **H04B7/06** (20060101); **H04W72/12** (20230101); **H04W72/54** (20230101)

U.S. Cl.:

CPC **H04B17/345** (20150115); **H04B7/0632** (20130101); **H04W72/12** (20130101); **H04W72/54** (20230101);

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION(S) [0001] This application is a Continuation-In-Part application of U.S. patent application Ser. No. 18/592,389, entitled “ENHANCED UE MAXIMUM SENSITIVITY DEGRADATION INDICATION” and filed on Feb. 29, 2024, which is expressly incorporated by reference herein in their entirety.

TECHNICAL FIELD

[0002] The present disclosure relates generally to communication systems, and more particularly, to a configuration for an enhanced user equipment (UE) maximum sensitivity degradation indication.

INTRODUCTION

[0003] Wireless communication systems are widely deployed to provide various telecommunication services such as telephony, video, data, messaging, and broadcasts. Typical wireless communication systems may employ multiple-access technologies capable of supporting communication with multiple users by sharing available system resources. Examples of such multiple-access technologies include code division multiple access (CDMA) systems, time division multiple access (TDMA) systems, frequency division multiple access (FDMA) systems, orthogonal frequency division multiple access (OFDMA) systems, single-carrier frequency division multiple access (SC-FDMA) systems, and time division synchronous code division multiple access (TD-SCDMA) systems.

[0004] These multiple access technologies have been adopted in various telecommunication standards to provide a common protocol that enables different wireless devices to communicate on a municipal, national, regional, and even global level. An example telecommunication standard is 5G New Radio (NR). 5G NR is part of a continuous mobile broadband evolution promulgated by Third Generation Partnership Project (3GPP) to meet new requirements associated with latency, reliability, security, scalability (e.g., with Internet of Things (IoT)), and other requirements. 5G NR includes services associated with enhanced mobile broadband (eMBB), massive machine type communications (mMTC), and ultra-reliable low latency communications (URLLC). Some aspects of 5G NR may be based on the 4G Long Term Evolution (LTE) standard. There exists a need for further improvements in 5G NR technology. These improvements may also be applicable to other multi-access technologies and the telecommunication standards that employ these technologies.

BRIEF SUMMARY

[0005] The following presents a simplified summary of one or more aspects in order to provide a basic understanding of such aspects. This summary is not an extensive overview of all contemplated aspects. This summary neither identifies key or critical elements of all aspects nor delineates the scope of any or all aspects. Its sole purpose is to present some concepts of one or more aspects in a simplified form as a prelude to the more detailed description that is presented later.

[0006] In an aspect of the disclosure, a method, a computer-readable medium, and an apparatus are provided. The apparatus may be a device at a user equipment (UE). The device may be a processor and/or a modem at a UE or the UE itself. The apparatus receives a request for a measurement of a maximum sensitivity degradation (MSD). The apparatus measures the MSD caused by self-interference or a different interference than the self-interference. The apparatus reports a report comprising the measurement of the MSD including an interference source indication that the MSD is due to the self-interference, the different interference, or a combination of the self-interference and the different interference.

[0007] In an aspect of the disclosure, a method, a computer-readable medium, and an apparatus are provided. The apparatus may be a device at a UE. The device may be a processor and/or a modem

at a UE or the UE itself. The apparatus receives a request for a measurement of an MSD. The apparatus measures the MSD caused by self-interference or a different interference than the self-interference. The apparatus reports a report comprising the measurement of the MSD including relative information associated with a first amount of the MSD caused by the self-interference.
[0008] In an aspect of the disclosure, a method, a computer-readable medium, and an apparatus are provided. The apparatus may be a device at a network node. The device may be a processor and/or a modem at a network node or the network node itself. The apparatus provides a request for a measurement of a maximum sensitivity degradation (MSD). The apparatus obtains a report comprising the measurement of the MSD caused by self-interference at a user equipment (UE) or a different interference than the self-interference at the UE, wherein the report includes an interference source indication that the MSD is due to the self-interference, the different interference, or a combination of the self-interference and the different interference. The apparatus schedules communication based on the MSD.
[0009] In an aspect of the disclosure, a method, a computer-readable medium, and an apparatus are provided. The apparatus may be a device at a network node. The device may be a processor and/or a modem at a network node or the network node itself. The apparatus provides a request for a measurement of an MSD. The apparatus obtains a report comprising the measurement of the MSD caused by self-interference at a UE or a different interference than the self-interference at the UE, wherein the report includes relative information associated with a first amount of the MSD caused by the self-interference. The apparatus schedules communication based on the MSD.
[0010] To the accomplishment of the foregoing and related ends, the one or more aspects may include the features hereinafter fully described and particularly pointed out in the claims. The following description and the drawings set forth in detail certain illustrative features of the one or more aspects. These features are indicative, however, of but a few of the various ways in which the principles of various aspects may be employed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0011] FIG. **1** is a diagram illustrating an example of a wireless communications system and an access network.
[0012] FIG. **2**A is a diagram illustrating an example of a first frame, in accordance with various aspects of the present disclosure.
[0013] FIG. **2**B is a diagram illustrating an example of downlink (DL) channels within a subframe, in accordance with various aspects of the present disclosure.
[0014] FIG. **2**C is a diagram illustrating an example of a second frame, in accordance with various aspects of the present disclosure.
[0015] FIG. **2**D is a diagram illustrating an example of uplink (UL) channels within a subframe, in accordance with various aspects of the present disclosure.
[0016] FIG. **3** is a diagram illustrating an example of a base station and user equipment (UE) in an access network.
[0017] FIG. **4** is a diagram illustrating an example of self-interference at a UE.
[0018] FIG. **5**A is a diagram illustrating an example of a UE.
[0019] FIG. **5**B is a diagram illustrating an example of a UE.
[0020] FIG. **6** is a call flow diagram of signaling between a UE and a base station.
[0021] FIG. **7** is a flowchart of a method of wireless communication.
[0022] FIG. **8** is a flowchart of a method of wireless communication.
[0023] FIG. **9** is a diagram illustrating an example of a hardware implementation for an example apparatus and/or network entity.

[0024] FIG. **10** is a flowchart of a method of wireless communication.
[0025] FIG. **11** is a flowchart of a method of wireless communication.
[0026] FIG. **12** is a diagram illustrating an example of a hardware implementation for an example network entity.

DETAILED DESCRIPTION

[0027] In wireless communication, maximum sensitivity degradation (MSD) is the degradation of DL reference sensitivity (REFSENS) in carrier aggregation (CA) or Evolved-Universal Terrestrial Radio Access-New Radio Dual Connectivity (EN-DC) compared to non-CA/non-EN-DC REFSENS. MSD may be caused by self-interference due to certain frequency relation of the frequency bands, channels, or RB allocations (e.g., tx harmonics, rx mixing, intermodulations, cross-band, etc.). MSD may be specified in conductive conditions, such that antenna characteristics are virtually excluded or in instances where a hand or other body part of a user blocks the device which impacts antenna gains are not taken into consideration. Over-the-air (OTA) MSD may be caused by interference, while sensitivity may be degraded by other factors, such as but not limited to a user hand grip on the device or the like, which results in a reduced OTA REFSENS in comparison to free-space REFSENS even when there is no MSD caused by self-interference. In some instances, sensitivity degradation caused by other factors (e.g user hand grip on the device or other mechanisms) may exclude self-interference and may be referred to as other or different interference.
[0028] A UE OTA REFSENS may vary several dBs depending on the manner in which the device is held and depending on whether there is self-interference due to certain frequency relations between band combinations. In some instances, for example, if a UE indicates an MSD of 10 dB due to self-interference, the network may release or not configure that configuration because MSD is considered high. However, if that same REFSENS degradation is due to the user hand grip on the device (e.g., other or different interference), then the network may not release that configuration since a user holding device in their hand is commonplace. As such, it would be advantageous to distinguish these two sources of degradation in rigorous MSD indication.
[0029] Aspects presented herein provide a configuration for an enhanced UE MSD indication, such that the UE may indicate the MSD that is caused by self-interference or by other or different interference. The UE may indicate the total MSD based on self-interference, other or different interference, or a combination of the self-interference and the other of different interference, as well as indicate which degradation source is a dominant source, or the amount of degradation due to the self-interference or the other or different interference.
[0030] The detailed description set forth below in connection with the drawings describes various configurations and does not represent the only configurations in which the concepts described herein may be practiced. The detailed description includes specific details for the purpose of providing a thorough understanding of various concepts. However, these concepts may be practiced without these specific details. In some instances, well known structures and components are shown in block diagram form in order to avoid obscuring such concepts.
[0031] Several aspects of telecommunication systems are presented with reference to various apparatus and methods. These apparatus and methods are described in the following detailed description and illustrated in the accompanying drawings by various blocks, components, circuits, processes, algorithms, etc. (collectively referred to as “elements”). These elements may be implemented using electronic hardware, computer software, or any combination thereof. Whether such elements are implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system.
[0032] By way of example, an element, or any portion of an element, or any combination of elements may be implemented as a “processing system” that includes one or more processors. When multiple processors are implemented, the multiple processors may perform the functions individually or in combination. Examples of processors include microprocessors, microcontrollers,

graphics processing units (GPUs), central processing units (CPUs), application processors, digital signal processors (DSPs), reduced instruction set computing (RISC) processors, systems on a chip (SoC), baseband processors, field programmable gate arrays (FPGAs), programmable logic devices (PLDs), state machines, gated logic, discrete hardware circuits, and other suitable hardware configured to perform the various functionality described throughout this disclosure. One or more processors in the processing system may execute software. Software, whether referred to as software, firmware, middleware, microcode, hardware description language, or otherwise, shall be construed broadly to mean instructions, instruction sets, code, code segments, program code, programs, subprograms, software components, applications, software applications, software packages, routines, subroutines, objects, executables, threads of execution, procedures, functions, or any combination thereof.

[0033] Accordingly, in one or more example aspects, implementations, and/or use cases, the functions described may be implemented in hardware, software, or any combination thereof. If implemented in software, the functions may be stored on or encoded as one or more instructions or code on a computer-readable medium. Computer-readable media includes computer storage media. Storage media may be any available media that can be accessed by a computer. By way of example, such computer-readable media can include a random-access memory (RAM), a read-only memory (ROM), an electrically erasable programmable ROM (EEPROM), optical disk storage, magnetic disk storage, other magnetic storage devices, combinations of the types of computer-readable media, or any other medium that can be used to store computer executable code in the form of instructions or data structures that can be accessed by a computer. While aspects, implementations, and/or use cases are described in this application by illustration to some examples, additional or different aspects, implementations and/or use cases may come about in many different arrangements and scenarios. Aspects, implementations, and/or use cases described herein may be implemented across many differing platform types, devices, systems, shapes, sizes, and packaging arrangements. For example, aspects, implementations, and/or use cases may come about via integrated chip implementations and other non-module-component based devices (e.g., end-user devices, vehicles, communication devices, computing devices, industrial equipment, retail/purchasing devices, medical devices, artificial intelligence (AI)-enabled devices, etc.). While some examples may or may not be specifically directed to use cases or applications, a wide assortment of applicability of described examples may occur. Aspects, implementations, and/or use cases may range a spectrum from chip-level or modular components to non-modular, non-chip-level implementations and further to aggregate, distributed, or original equipment manufacturer (OEM) devices or systems incorporating one or more techniques herein. In some practical settings, devices incorporating described aspects and features may also include additional components and features for implementation and practice of claimed and described aspect. For example, transmission and reception of wireless signals necessarily includes a number of components for analog and digital purposes (e.g., hardware components including antenna, RF-chains, power amplifiers, modulators, buffer, processor(s), interleaver, adders/summers, etc.). Techniques described herein may be practiced in a wide variety of devices, chip-level components, systems, distributed arrangements, aggregated or disaggregated components, end-user devices, etc. of varying sizes, shapes, and constitution.

[0034] Deployment of communication systems, such as 5G NR systems, may be arranged in multiple manners with various components or constituent parts. In a 5G NR system, or network, a network node, a network entity, a mobility element of a network, a radio access network (RAN) node, a core network node, a network element, or a network equipment, such as a base station (BS), or one or more units (or one or more components) performing base station functionality, may be implemented in an aggregated or disaggregated architecture. For example, a BS (such as a Node B (NB), evolved NB (eNB), NR BS, 5G NB, access point (AP), a transmission reception point (TRP), or a cell, etc.) may be implemented as an aggregated base station (also known as a standalone BS or a monolithic BS) or a disaggregated base station.

[0035] An aggregated base station may be configured to utilize a radio protocol stack that is physically or logically integrated within a single RAN node. A disaggregated base station may be configured to utilize a protocol stack that is physically or logically distributed among two or more units (such as one or more central or centralized units (CUs), one or more distributed units (DUs), or one or more radio units (RUs)). In some aspects, a CU may be implemented within a RAN node, and one or more DUs may be co-located with the CU, or alternatively, may be geographically or virtually distributed throughout one or multiple other RAN nodes. The DUs may be implemented to communicate with one or more RUs. Each of the CU, DU and RU can be implemented as virtual units, i.e., a virtual central unit (VCU), a virtual distributed unit (VDU), or a virtual radio unit (VRU).

[0036] Base station operation or network design may consider aggregation characteristics of base station functionality. For example, disaggregated base stations may be utilized in an integrated access backhaul (IAB) network, an open radio access network (O-RAN (such as the network configuration sponsored by the O-RAN Alliance)), or a virtualized radio access network (vRAN, also known as a cloud radio access network (C-RAN)). Disaggregation may include distributing functionality across two or more units at various physical locations, as well as distributing functionality for at least one unit virtually, which can enable flexibility in network design. The various units of the disaggregated base station, or disaggregated RAN architecture, can be configured for wired or wireless communication with at least one other unit.

[0037] FIG. **1** is a diagram **100** illustrating an example of a wireless communications system and an access network. The illustrated wireless communications system includes a disaggregated base station architecture. The disaggregated base station architecture may include one or more CUs **110** that can communicate directly with a core network **120** via a backhaul link, or indirectly with the core network **120** through one or more disaggregated base station units (such as a Near-Real Time (Near-RT) RAN Intelligent Controller (RIC) **125** via an E2 link, or a Non-Real Time (Non-RT) RIC **115** associated with a Service Management and Orchestration (SMO) Framework **105**, or both). A CU **110** may communicate with one or more DUs **130** via respective midhaul links, such as an F1 interface. The DUs **130** may communicate with one or more RUs **140** via respective fronthaul links. The RUs **140** may communicate with respective UEs **104** via one or more radio frequency (RF) access links. In some implementations, the UE **104** may be simultaneously served by multiple RUs **140**.

[0038] Each of the units, i.e., the CUs **110**, the DUs **130**, the RUs **140**, as well as the Near-RT RICs **125**, the Non-RT RICs **115**, and the SMO Framework **105**, may include one or more interfaces or be coupled to one or more interfaces configured to receive or to transmit signals, data, or information (collectively, signals) via a wired or wireless transmission medium. Each of the units, or an associated processor or controller providing instructions to the communication interfaces of the units, can be configured to communicate with one or more of the other units via the transmission medium. For example, the units can include a wired interface configured to receive or to transmit signals over a wired transmission medium to one or more of the other units. Additionally, the units can include a wireless interface, which may include a receiver, a transmitter, or a transceiver (such as an RF transceiver), configured to receive or to transmit signals, or both, over a wireless transmission medium to one or more of the other units.

[0039] In some aspects, the CU **110** may host one or more higher layer control functions. Such control functions can include radio resource control (RRC), packet data convergence protocol (PDCP), service data adaptation protocol (SDAP), or the like. Each control function can be implemented with an interface configured to communicate signals with other control functions hosted by the CU **110**. The CU **110** may be configured to handle user plane functionality (i.e., Central Unit-User Plane (CU-UP)), control plane functionality (i.e., Central Unit-Control Plane (CU-CP)), or a combination thereof. In some implementations, the CU **110** can be logically split into one or more CU-UP units and one or more CU-CP units. The CU-UP unit can communicate bidirectionally with the CU-CP unit via an interface, such as an E1 interface when implemented in an O-RAN configuration. The CU **110** can be implemented to communicate with the DU **130**, as necessary, for network control and signaling.

[0040] The DU **130** may correspond to a logical unit that includes one or more base station functions to control the operation of one or more RUs **140**. In some aspects, the DU **130** may host one or more of a radio link control (RLC) layer, a medium access control (MAC) layer, and one or more high physical (PHY) layers (such as modules for forward error correction (FEC) encoding and decoding, scrambling, modulation, demodulation, or the like) depending, at least in part, on a functional split, such as those defined by 3GPP. In some aspects, the DU **130** may further host one or more low PHY layers. Each layer (or module) can be implemented with an interface configured to communicate signals with other layers (and modules) hosted by the DU **130**, or with the control functions hosted by the CU **110**.

[0041] Lower-layer functionality can be implemented by one or more RUs **140**. In some deployments, an RU **140**, controlled by a DU **130**, may correspond to a logical node that hosts RF processing functions, or low-PHY layer functions (such as performing fast Fourier transform (FFT), inverse FFT (iFFT), digital beamforming, physical random access channel (PRACH) extraction and filtering, or the like), or both, based at least in part on the functional split, such as a lower layer functional split. In such an architecture, the RU(s) **140** can be implemented to handle over the air (OTA) communication with one or more UEs **104**. In some implementations, real-time and non-real-time aspects of control and user plane communication with the RU(s) **140** can be controlled by the corresponding DU **130**. In some scenarios, this configuration can enable the DU(s) **130** and the CU **110** to be implemented in a cloud-based RAN architecture, such as a vRAN architecture.

[0042] The SMO Framework **105** may be configured to support RAN deployment and provisioning of non-virtualized and virtualized network elements. For non-virtualized network elements, the SMO Framework **105** may be configured to support the deployment of dedicated physical resources for RAN coverage requirements that may be managed via an operations and maintenance interface (such as an O1 interface). For virtualized network elements, the SMO Framework **105** may be configured to interact with a cloud computing platform (such as an open cloud (O-Cloud) **190**) to perform network element life cycle management (such as to instantiate virtualized network elements) via a cloud computing platform interface (such as an O2 interface). Such virtualized network elements can include, but are not limited to, CUs **110**, DUs **130**, RUs **140** and Near-RT RICs **125**. In some implementations, the SMO Framework **105** can communicate with a hardware aspect of a 4G RAN, such as an open eNB (O-eNB) **111**, via an O1 interface. Additionally, in some implementations, the SMO Framework **105** can communicate directly with one or more RUs **140** via an O1 interface. The SMO Framework **105** also may include a Non-RT RIC **115** configured to support functionality of the SMO Framework **105**.

[0043] The Non-RT RIC **115** may be configured to include a logical function that enables non-real-time control and optimization of RAN elements and resources, artificial intelligence (AI)/machine learning (ML) (AI/ML) workflows including model training and updates, or policy-based guidance of applications/features in the Near-RT RIC **125**. The Non-RT RIC **115** may be coupled to or communicate with (such as via an A1 interface) the Near-RT RIC **125**. The Near-RT RIC **125** may be configured to include a logical function that enables near-real-time control and optimization of RAN elements and resources via data collection and actions over an interface (such as via an E2 interface) connecting one or more CUs **110**, one or more DUs **130**, or both, as well as an O-eNB, with the Near-RT RIC **125**.

[0044] In some implementations, to generate AI/ML models to be deployed in the Near-RT RIC **125**, the Non-RT RIC **115** may receive parameters or external enrichment information from external servers. Such information may be utilized by the Near-RT RIC **125** and may be received at the SMO Framework **105** or the Non-RT RIC **115** from non-network data sources or from network functions. In some examples, the Non-RT RIC **115** or the Near-RT RIC **125** may be configured to tune RAN behavior or performance. For example, the Non-RT RIC **115** may monitor long-term trends and patterns for performance and employ AI/ML models to perform corrective actions through the SMO Framework **105** (such as reconfiguration via **01**) or via creation of RAN management policies (such as A1 policies).

[0045] At least one of the CU **110**, the DU **130**, and the RU **140** may be referred to as a base station **102**. Accordingly, a base station **102** may include one or more of the CU **110**, the DU **130**, and the RU **140** (each component indicated with dotted lines to signify that each component may or may not be included in the base station **102**). The base station **102** provides an access point to the core network **120** for a UE **104**. The base station **102** may include macrocells (high power cellular base station) and/or small cells (low power cellular base station). The small cells include femtocells, picocells, and microcells. A network that includes both small cell and macrocells may be known as a heterogeneous network. A heterogeneous network may also include Home Evolved Node Bs (eNBs) (HeNBs), which may provide service to a restricted group known as a closed subscriber group (CSG). The communication links between the RUs **140** and the UEs **104** may include uplink (UL) (also referred to as reverse link) transmissions from a UE **104** to an RU **140** and/or downlink (DL) (also referred to as forward link) transmissions from an RU **140** to a UE **104**. The communication links may use multiple-input and multiple-output (MIMO) antenna technology, including spatial multiplexing, beamforming, and/or transmit diversity. The communication links may be through one or more carriers. The base station **102**/UEs **104** may use spectrum up to Y MHz (e.g., 5, 10, 15, 20, 100, 400, etc. MHz) bandwidth per carrier allocated in a carrier aggregation of up to a total of Yx MHz (x component carriers) used for transmission in each direction. The carriers may or may not be adjacent to each other. Allocation of carriers may be asymmetric with respect to DL and UL (e.g., more or fewer carriers may be allocated for DL than for UL). The component carriers may include a primary component carrier and one or more secondary component carriers. A primary component carrier may be referred to as a primary cell (PCell) and a secondary component carrier may be referred to as a secondary cell (SCell).

[0046] Certain UEs **104** may communicate with each other using device-to-device (D2D) communication link **158**. The D2D communication link **158** may use the DL/UL wireless wide area network (WWAN) spectrum. The D2D communication link **158** may use one or more sidelink channels, such as a physical sidelink broadcast channel (PSBCH), a physical sidelink discovery channel (PSDCH), a physical sidelink shared channel (PSSCH), and a physical sidelink control channel (PSCCH). D2D communication may be through a variety of wireless D2D communications systems, such as for example, Bluetooth™ (Bluetooth is a trademark of the Bluetooth Special Interest Group (SIG)), Wi-Fi™ (Wi-Fi is a trademark of the Wi-Fi Alliance) based on the Institute of Electrical and Electronics Engineers (IEEE) 802.11 standard, LTE, or NR.

[0047] The wireless communications system may further include a Wi-Fi AP **150** in communication with UEs **104** (also referred to as Wi-Fi stations (STAs)) via communication link **154**, e.g., in a 5 GHz unlicensed frequency spectrum or the like. When communicating in an unlicensed frequency spectrum, the UEs **104**/AP **150** may perform a clear channel assessment (CCA) prior to communicating in order to determine whether the channel is available.

[0048] The electromagnetic spectrum is often subdivided, based on frequency/wavelength, into various classes, bands, channels, etc. In 5G NR, two initial operating bands have been identified as frequency range designations FR1 (410 MHz-7.125 GHZ) and FR2 (24.25 GHz-52.6 GHz). Although a portion of FR1 is greater than 6 GHz, FR1 is often referred to (interchangeably) as a “sub-6 GHz” band in various documents and articles. A similar nomenclature issue sometimes occurs with regard to FR2, which is often referred to (interchangeably) as a “millimeter wave” band in documents and articles, despite being different from the extremely high frequency (EHF) band (30 GHz-300 GHz) which is identified by the International Telecommunications Union (ITU) as a "millimeter wave" band.

[0049] The frequencies between FR1 and FR2 are often referred to as mid-band frequencies. Recent 5G NR studies have identified an operating band for these mid-band frequencies as frequency range designation FR3 (7.125 GHZ-24.25 GHZ). Frequency bands falling within FR3 may inherit FR1 characteristics and/or FR2 characteristics, and thus may effectively extend features of FR1 and/or FR2 into mid-band frequencies. In addition, higher frequency bands are currently being explored to extend 5G NR operation beyond 52.6 GHz. For example, three higher operating bands have been identified as frequency range designations FR2-2 (52.6 GHz-71 GHZ), FR4 (71 GHz-114.25 GHz), and FR5 (114.25 GHz-300 GHz). Each of these higher frequency bands falls within the EHF band.

[0050] With the above aspects in mind, unless specifically stated otherwise, the term "sub-6 GHz" or the like if used herein may broadly represent frequencies that may be less than 6 GHz, may be within FR1, or may include mid-band frequencies. Further, unless specifically stated otherwise, the term "millimeter wave" or the like if used herein may broadly represent frequencies that may include mid-band frequencies, may be within FR2, FR4, FR2-2, and/or FR5, or may be within the EHF band.

[0051] The base station **102** and the UE **104** may each include a plurality of antennas, such as antenna elements, antenna panels, and/or antenna arrays to facilitate beamforming. The base station **102** may transmit a beamformed signal **182** to the UE **104** in one or more transmit directions. The UE **104** may receive the beamformed signal from the base station **102** in one or more receive directions. The UE **104** may also transmit a beamformed signal **184** to the base station **102** in one or more transmit directions. The base station **102** may receive the beamformed signal from the UE **104** in one or more receive directions. The base station **102**/UE **104** may perform beam training to determine the best receive and transmit directions for each of the base station **102**/UE **104**. The transmit and receive directions for the base station **102** may or may not be the same. The transmit and receive directions for the UE **104** may or may not be the same.

[0052] The base station **102** may include and/or be referred to as a gNB, Node B, eNB, an access point, a base transceiver station, a radio base station, a radio transceiver, a transceiver function, a basic service set (BSS), an extended service set (ESS), a TRP, network node, network entity, network equipment, or some other suitable terminology. The base station **102** can be implemented as an integrated access and backhaul (IAB) node, a relay node, a sidelink node, an aggregated (monolithic) base station with a baseband unit (BBU) (including a CU and a DU) and an RU, or as a disaggregated base station including one or more of a CU, a DU, and/or an RU. The set of base stations, which may include disaggregated base stations and/or aggregated base stations, may be referred to as next generation (NG) RAN (NG-RAN).

[0053] The core network **120** may include an Access and Mobility Management Function (AMF) **161**, a Session Management Function (SMF) **162**, a User Plane Function (UPF) **163**, a Unified Data Management (UDM) **164**, one or more location servers **168**, and other functional entities. The AMF **161** is the control node that processes the signaling between the UEs **104** and the core network **120**. The AMF **161** supports registration management, connection management, mobility management, and other functions. The SMF **162** supports session management and other functions. The UPF **163** supports packet routing, packet forwarding, and other functions. The UDM **164** supports the generation of authentication and key agreement (AKA) credentials, user identification handling, access authorization, and subscription management. The one or more location servers **168** are illustrated as including a Gateway Mobile Location Center (GMLC) **165** and a Location Management Function (LMF) **166**. However, generally, the one or more location servers **168** may include one or more location/positioning servers, which may include one or more of the GMLC **165**, the LMF **166**, a position determination entity (PDE), a serving mobile location center (SMLC), a mobile positioning center (MPC), or the like. The GMLC **165** and the LMF **166** support UE location services. The GMLC **165** provides an interface for clients/applications (e.g., emergency services) for accessing UE positioning information. The LMF **166** receives measurements and assistance information from the NG-RAN and the UE **104** via the AMF **161** to compute the position of the UE **104**. The NG-RAN may utilize one or more positioning methods in order to determine the position of the UE **104**. Positioning the UE **104** may involve signal measurements, a position estimate, and an optional velocity computation based on the measurements. The signal measurements may be made by the UE **104** and/or the base station **102** serving the UE **104**. The signals measured may be based on one or more of a satellite positioning system (SPS) **170** (e.g., one or more of a Global Navigation Satellite System (GNSS), global position system (GPS), non-terrestrial network (NTN), or other satellite position/location system), LTE signals, wireless local area network (WLAN) signals, Bluetooth signals, a terrestrial beacon system (TBS), sensor-based information (e.g., barometric pressure sensor, motion sensor), NR enhanced cell ID (NR E-CID) methods, NR signals (e.g., multi-round trip time (Multi-RTT), DL angle-of-departure (DL-AoD), DL time difference of arrival (DL-TDOA), UL time difference of arrival (UL-TDOA), and UL angle-of-arrival (UL-AoA) positioning), and/or other systems/signals/sensors.

[0054] Examples of UEs **104** include a cellular phone, a smart phone, a session initiation protocol (SIP) phone, a laptop, a personal digital assistant (PDA), a satellite radio, a global positioning system, a multimedia device, a video device, a digital audio player (e.g., MP3 player), a camera, a game console, a tablet, a smart device, a wearable device, a vehicle, an electric meter, a gas pump, a large or small kitchen appliance, a healthcare device, an implant, a sensor/actuator, a display, or any other similar functioning device. Some of the UEs **104** may be referred to as IoT devices (e.g., parking meter, gas pump, toaster, vehicles, heart monitor, etc.). The UE **104** may also be referred to as a station, a mobile station, a subscriber station, a mobile unit, a subscriber unit, a wireless unit, a remote unit, a mobile device, a wireless device, a wireless communications device, a remote device, a mobile subscriber station, an access terminal, a mobile terminal, a wireless terminal, a remote terminal, a handset, a user agent, a mobile client, a client, or some other suitable terminology. In some scenarios, the term UE may also apply to one or more companion devices such as in a device constellation arrangement. One or more of these devices may collectively access the network and/or individually access the network.

[0055] Referring again to FIG. **1**, in certain aspects, the UE **104** may include a measurement component **198** that may be configured to receive a request for a measurement of an MSD; measure the MSD caused by self-interference or a different interference than the self-interference; and report a report comprising the measurement of the MSD including an interference source indication that the MSD is due to the self-interference, the different interference, or a combination of the self-interference and the different interference.

[0056] Referring again to FIG. **1**, in certain aspects, the base station **102** may include a measurement component **199** that may be configured to provide a request for a measurement of an MSD; obtain a report comprising the measurement of the MSD caused by self-interference at a UE or a different interference than the self-interference at the UE, wherein the report includes an interference source indication that the MSD is due to the self-interference, the different interference, or a combination of the self-interference and the different interference; and schedule communication based on the MSD.

[0057] Although the following description may be focused on 5G NR, the concepts described herein may be applicable to other similar areas, such as LTE, LTE-A, CDMA, GSM, and other wireless technologies.

[0058] FIG. **2**A is a diagram **200** illustrating an example of a first subframe within a 5G NR frame structure. FIG. **2**B is a diagram **230** illustrating an example of DL channels within a 5G NR subframe. FIG. **2**C is a diagram **250** illustrating an example of a second subframe within a 5G NR frame structure. FIG. **2**D is a diagram **280** illustrating an example of UL channels within a 5G NR subframe. The 5G NR frame structure may be frequency division duplexed (FDD) in which for a particular set of subcarriers (carrier system bandwidth), subframes within the set of subcarriers are dedicated for either DL or UL, or may be time division duplexed (TDD) in which for a particular set of subcarriers (carrier system bandwidth), subframes within the set of subcarriers are dedicated for both DL and UL. In the examples provided by FIGS. **2**A, **2**C, the 5G NR frame structure is assumed to be TDD, with subframe 4 being configured with slot format 28 (with mostly DL), where D is DL, U is UL, and F is flexible for use between DL/UL, and subframe 3 being configured with slot format 1 (with all UL). While subframes 3, 4 are shown with slot formats 1, 28, respectively, any particular subframe may be configured with any of the various available slot formats 0-61. Slot formats 0, 1 are all DL, UL, respectively. Other slot formats 2-61 include a mix of DL, UL, and flexible symbols. UEs are configured with the slot format (dynamically through DL control information (DCI), or semi-statically/statically through radio resource control (RRC) signaling) through a received slot format indicator (SFI). Note that the description infra applies also to a 5G NR frame structure that is TDD.

[0059] FIGS. **2**A-**2**D illustrate a frame structure, and the aspects of the present disclosure may be applicable to other wireless communication technologies, which may have a different frame structure and/or different channels. A frame (10 ms) may be divided into 10 equally sized subframes (1 ms). Each subframe may include one or more time slots. Subframes may also include mini-slots, which may include 7, 4, or 2 symbols. Each slot may include 14 or 12 symbols, depending on whether the cyclic prefix (CP) is normal or extended. For normal CP, each slot may include 14 symbols, and for extended CP, each slot may include 12 symbols. The symbols on DL may be CP orthogonal frequency division multiplexing (OFDM) (CP-OFDM) symbols. The symbols on UL may be CP-OFDM symbols (for high throughput scenarios) or discrete Fourier transform (DFT) spread OFDM (DFT-s-OFDM) symbols (for power limited scenarios; limited to a single stream transmission). The number of slots within a subframe is based on the CP and the numerology. The numerology defines the subcarrier spacing (SCS) (see Table 1). The symbol length/duration may scale with 1/SCS.

TABLE-US-00001 TABLE 1 Numerology, SCS, and CP SCS μ Δf = 2.sup.μ .Math. 15[KHz] Cyclic prefix 0 15 Normal 1 30 Normal 2 60 Normal, Extended 3 120 Normal 4 240 Normal 5 480 Normal 6 960 Normal

[0060] For normal CP (14 symbols/slot), different numerologies μ 0 to 4 allow for 1, 2, 4, 8, and 16 slots, respectively, per subframe. For extended CP, the numerology 2 allows for 4 slots per subframe. Accordingly, for normal CP and numerology μ, there are 14 symbols/slot and 2.sup.μ slots/subframe. The subcarrier spacing may be equal to 2.sup.μ*15 kHz, where μ is the numerology 0 to 4. As such, the numerology μ=0 has a subcarrier spacing of 15 kHz and the numerology μ=4 has a subcarrier spacing of 240 kHz. The symbol length/duration is inversely related to the subcarrier spacing. FIGS. **2**A-**2**D provide an example of normal CP with 14 symbols per slot and numerology μ=2 with 4 slots per subframe. The slot duration is 0.25 ms, the subcarrier spacing is 60 kHz, and the symbol duration is approximately 16.67 μs. Within a set of frames, there may be one or more different bandwidth parts (BWPs) (see FIG. **2**B) that are frequency division multiplexed. Each BWP may have a particular numerology and CP (normal or extended).

[0061] A resource grid may be used to represent the frame structure. Each time slot includes a resource block (RB) (also referred to as physical RBs (PRBs)) that extends 12 consecutive subcarriers. The resource grid is divided into multiple resource elements (REs). The number of bits carried by each RE depends on the modulation scheme.

[0062] As illustrated in FIG. **2**A, some of the REs carry reference (pilot) signals (RS) for the UE. The RS may include demodulation RS (DM-RS) (indicated as R for one particular configuration, but other DM-RS configurations are possible) and channel state information reference signals (CSI-RS) for channel estimation at the UE. The RS may also include beam measurement RS (BRS), beam refinement RS (BRRS), and phase tracking RS (PT-RS).

[0063] FIG. **2**B illustrates an example of various DL channels within a subframe of a frame. The physical downlink control channel (PDCCH) carries DCI within one or more control channel elements (CCEs) (e.g., 1, 2, 4, 8, or 16 CCEs), each CCE including six RE groups (REGs), each REG including 12 consecutive REs in an OFDM symbol of an RB. A PDCCH within one BWP may be referred to as a control resource set (CORESET). A UE is configured to monitor PDCCH candidates in a PDCCH search space (e.g., common search space, UE-specific search space) during PDCCH monitoring occasions on the CORESET, where the PDCCH candidates have different DCI formats and different aggregation levels. Additional BWPs may be located at greater and/or lower frequencies across the channel bandwidth. A primary synchronization signal (PSS) may be within symbol 2 of particular subframes of a frame. The PSS is used by a UE **104** to determine subframe/symbol timing and a physical layer identity. A secondary synchronization signal (SSS) may be within symbol 4 of particular subframes of a frame. The SSS is used by a UE to determine a physical layer cell identity group number and radio frame timing. Based on the physical layer identity and the physical layer cell identity group number, the UE can determine a physical cell identifier (PCI). Based on the PCI, the UE can determine the locations of the DM-RS. The physical broadcast channel (PBCH), which carries a master information block (MIB), may be logically grouped with the PSS and SSS to form a synchronization signal (SS)/PBCH block (also referred to as SS block (SSB)). The MIB provides a number of RBs in the system bandwidth and a system frame number (SFN). The physical downlink shared channel (PDSCH) carries user data, broadcast system information not transmitted through the PBCH such as system information blocks (SIBs), and paging messages.

[0064] As illustrated in FIG. **2**C, some of the REs carry DM-RS (indicated as R for one particular configuration, but other DM-RS configurations are possible) for channel estimation at the base station. The UE may transmit DM-RS for the physical uplink control channel (PUCCH) and DM-RS for the physical uplink shared channel (PUSCH). The PUSCH DM-RS may be transmitted in the first one or two symbols of the PUSCH. The PUCCH DM-RS may be transmitted in different configurations depending on whether short or long PUCCHs are transmitted and depending on the particular PUCCH format used. The UE may transmit sounding reference signals (SRS). The SRS may be transmitted in the last symbol of a subframe. The SRS may have a comb structure, and a UE may transmit SRS on one of the combs. The SRS may be used by a base station for channel quality estimation to enable frequency-dependent scheduling on the UL.

[0065] FIG. **2**D illustrates an example of various UL channels within a subframe of a frame. The PUCCH may be located as indicated in one configuration. The PUCCH carries uplink control information (UCI), such as scheduling requests, a channel quality indicator (CQI), a precoding matrix indicator (PMI), a rank indicator (RI), and hybrid automatic repeat request (HARQ) acknowledgment (ACK) (HARQ-ACK) feedback (i.e., one or more HARQ ACK bits indicating one or more ACK and/or negative ACK (NACK)). The PUSCH carries data, and may additionally be used to carry a buffer status report (BSR), a power headroom report (PHR), and/or UCI.

[0066] FIG. **3** is a block diagram of a base station **310** in communication with a UE **350** in an access network. In the DL, Internet protocol (IP) packets may be provided to a controller/processor **375**. The controller/processor **375** implements layer 3 and layer 2 functionality. Layer 3 includes a radio resource control (RRC) layer, and layer 2 includes a service data adaptation protocol (SDAP) layer, a packet data convergence protocol (PDCP) layer, a radio link control (RLC) layer, and a medium access control (MAC) layer. The controller/processor **375** provides RRC layer functionality associated with broadcasting of system information (e.g., MIB, SIBs), RRC connection control (e.g., RRC connection paging, RRC connection establishment, RRC connection modification, and RRC connection release), inter radio access technology (RAT) mobility, and measurement configuration for UE measurement reporting; PDCP layer functionality associated with header compression/decompression, security (ciphering, deciphering, integrity protection, integrity verification), and handover support functions; RLC layer functionality associated with the transfer of upper layer packet data units (PDUs), error correction through ARQ, concatenation, segmentation, and reassembly of RLC service data units (SDUs), re-segmentation of RLC data PDUs, and reordering of RLC data PDUs; and MAC layer functionality associated with mapping between logical channels and transport channels, multiplexing of MAC SDUs onto transport blocks (TBs), demultiplexing of MAC SDUs from TBs, scheduling information reporting, error correction through HARQ, priority handling, and logical channel prioritization.

[0067] The transmit (TX) processor **316** and the receive (RX) processor **370** implement layer 1 functionality associated with various signal processing functions. Layer 1, which includes a physical (PHY) layer, may include error detection on the transport channels, forward error correction (FEC) coding/decoding of the transport channels, interleaving, rate matching, mapping onto physical channels, modulation/demodulation of physical channels, and MIMO antenna processing. The TX processor **316** handles mapping to signal constellations based on various modulation schemes (e.g., binary phase-shift keying (BPSK), quadrature phase-shift keying (QPSK), M-phase-shift keying (M-PSK), M-quadrature amplitude modulation (M-QAM)). The coded and modulated symbols may then be split into parallel streams. Each stream may then be mapped to an OFDM subcarrier, multiplexed with a reference signal (e.g., pilot) in the time and/or frequency domain, and then combined together using an Inverse Fast Fourier Transform (IFFT) to produce a physical channel carrying a time domain OFDM symbol stream. The OFDM stream is spatially precoded to produce multiple spatial streams. Channel estimates from a channel estimator **374** may be used to determine the coding and modulation scheme, as well as for spatial processing. The channel estimate may be derived from a reference signal and/or channel condition feedback transmitted by the UE **350**. Each spatial stream may then be provided to a different antenna **320** via a separate transmitter **318**Tx. Each transmitter **318**Tx may modulate a radio frequency (RF) carrier with a respective spatial stream for transmission.

[0068] At the UE **350**, each receiver **354**Rx receives a signal through its respective antenna **352**. Each receiver **354**Rx recovers information modulated onto an RF carrier and provides the information to the receive (RX) processor **356**. The TX processor **368** and the RX processor **356** implement layer 1 functionality associated with various signal processing functions. The RX processor **356** may perform spatial processing on the information to recover any spatial streams destined for the UE **350**. If multiple spatial streams are destined for the UE **350**, they may be combined by the RX processor **356** into a single OFDM symbol stream. The RX processor **356** then converts the OFDM symbol stream from the time-domain to the frequency domain using a Fast Fourier Transform (FFT). The frequency domain signal includes a separate OFDM symbol stream for each subcarrier of the OFDM signal. The symbols on each subcarrier, and the reference signal, are recovered and demodulated by determining the most likely signal constellation points transmitted by the base station **310**. These soft decisions may be based on channel estimates computed by the channel estimator **358**. The soft decisions are then decoded and deinterleaved to recover the data and control signals that were originally transmitted by the base station **310** on the physical channel. The data and control signals are then provided to the controller/processor **359**, which implements layer 3 and layer 2 functionality.

[0069] The controller/processor **359** can be associated with at least one memory **360** that stores program codes and data. The at least one memory **360** may be referred to as a computer-readable medium. In the UL, the controller/processor **359** provides demultiplexing between transport and logical channels, packet reassembly, deciphering, header decompression, and control signal processing to recover IP packets. The controller/processor **359** is also responsible for error detection using an ACK and/or NACK protocol to support HARQ operations.

[0070] Similar to the functionality described in connection with the DL transmission by the base station **310**, the controller/processor **359** provides RRC layer functionality associated with system information (e.g., MIB, SIBs) acquisition, RRC connections, and measurement reporting; PDCP layer functionality associated with header compression/decompression, and security (ciphering, deciphering, integrity protection, integrity verification); RLC layer functionality associated with the transfer of upper layer PDUs, error correction through ARQ, concatenation, segmentation, and reassembly of RLC SDUs, re-segmentation of RLC data PDUs, and reordering of RLC data PDUs; and MAC layer functionality associated with mapping between logical channels and transport channels, multiplexing of MAC SDUs onto TBs, demultiplexing of MAC SDUs from TBs, scheduling information reporting, error correction through HARQ, priority handling, and logical channel prioritization.

[0071] Channel estimates derived by a channel estimator **358** from a reference signal or feedback transmitted by the base station **310** may be used by the TX processor **368** to select the appropriate coding and modulation schemes, and to facilitate spatial processing. The spatial streams generated by the TX processor **368** may be provided to different antenna **352** via separate transmitters **354**Tx. Each transmitter **354**Tx may modulate an RF carrier with a respective spatial stream for transmission.

[0072] The UL transmission is processed at the base station **310** in a manner similar to that described in connection with the receiver function at the UE **350**. Each receiver **318**Rx receives a signal through its respective antenna **320**. Each receiver **318**Rx recovers information modulated onto an RF carrier and provides the information to a RX processor **370**.

[0073] The controller/processor **375** can be associated with at least one memory **376** that stores program codes and data. The at least one memory **376** may be referred to as a computer-readable medium. In the UL, the controller/processor **375** provides demultiplexing between transport and logical channels, packet reassembly, deciphering, header decompression, control signal processing to recover IP packets. The controller/processor **375** is also responsible for error detection using an ACK and/or NACK protocol to support HARQ operations.

[0074] At least one of the TX processor **368**, the RX processor **356**, and the controller/processor **359** may be configured to perform aspects in connection with the measurement component **198** of FIG. **1**.

[0075] At least one of the TX processor **316**, the RX processor **370**, and the controller/processor **375** may be configured to perform aspects in connection with the measurement component **199** of FIG. **1**.

[0076] In wireless communication, MSD is the degradation of DL REFSENS in CA or EN-DC compared to non-CA/non-EN-DC REFSENS. MSD may be caused by self-interference due to certain frequency relation of the frequency bands, channels, or RB allocations (e.g., tx harmonics, rx mixing, intermodulations, cross-band, etc.). For example, as shown in diagram **400** of FIG. **4**, a UE **402** may transmit an uplink signal **404**, but the UE may detect self-interference **406** based on the uplink signal **404**. MSD may be specified in conductive conditions, such that antenna characteristics are virtually excluded or in instances where a hand or other body part of a user blocks the device which impacts antenna gains are not taken into consideration. For example, as shown in diagram **500** of FIG. **5**A or diagram **520** of FIG. **5**B, a user's hand **504** may provide blockage of part of the UE **502** while the UE is in use. OTA MSD may be caused by interference, while sensitivity may be degraded by other factors, such as but not limited to a user hand grip on the device or the like, which results in a reduced OTA REFSENS in comparison to free-space REFSENS even when there is no MSD caused by self-interference. In some instances, sensitivity degradation caused by other factors (e.g user hand grip on the device or other mechanisms) may exclude self-interference and may be referred to as other or different interference.

[0077] In some instances, MSD may be specified for a limited set of worst test points for band combinations. However, this may not address channel combination granularity and the worst case may not occur often in reality. Having a worst case specified may prevent operators from deploying certain band combinations due to poor expected performance.

[0078] In some instances, a UE may be configured to indicate an enhanced MSD as a static capability per band combination, but this may be only with respect to the worst test points, discussed above, but this may not address the real or perceived MSD per channel combination.

[0079] A UE OTA REFSENS may vary several dBs depending on the manner in which the device is held and depending on whether there is self-interference due to certain frequency relations between band combinations. In some instances, for example, if a UE indicates an MSD of 10 dB due to self-interference, the network may release or not configure that configuration because MSD is considered high. However, if that same REFSENS degradation is due to the user hand grip on the device (e.g., other or different interference), then the network may not release that configuration since a user holding device in their hand is commonplace. As such, it would be advantageous to distinguish these two sources of degradation in rigorous MSD indication.
[0080] Aspects presented herein provide a configuration for an enhanced UE MSD indication, such that the UE may indicate the MSD that is caused by self-interference or by other or different interference. The UE may indicate the total MSD based on self-interference, other or different interference, or a combination of the self-interference and the other of different interference, as well as indicate which degradation source is a dominant source, or the amount of degradation due to the self-interference or the other or different interference.
[0081] In some aspects, a UE may be configured with the capability to measure and report the MSD in OTA conditions in rigorous manner for each channel combination based on the existing UE conditions. The UE may report if self-interference or other interference, or combination of these two causes MSD. In some aspects, the UE may be configured to measure and report the MSA in OTA conditions for an UL and DL grant pair scheduled by the network.
[0082] In some aspects, a UE may be configured to provide a channel combination specific indication which may provide a more precise information to the network, such that the network may be better suited to make configuration/scheduling decisions. At least one advantage of the disclosure is that it is beneficial to distinguish cases when the network can have an impact to MSD (e.g by changing scheduling to have different UL or DL grants, etc.) and when it cannot (e.g when MSD is caused by hand grip). For example, if a UE indicates that the MSD is 5 dB is due to other or different interference, the network knows that it can safely use a similar grant in future. However, if the UE indicated that the MSD is 5 dB and is due to self-interference, then the network may consider using a different grant in future.
[0083] In some instances, a UE may be configured to determine if MSD is caused by self-interference or by other or different interference, or by combination of these two. In combinations without self-interference channel combinations, the MSD, if any, may be caused by other or different interference (e.g., user hand grip on the device or other body part blocking the device). In self-interference channel combinations, the UE may deduce if the MSD is caused by self-interference, or partially or entirely by other or different interference (e.g., user hand grip on the device or other body part blocking the device).
[0084] In some instances, a UE may be configured to indicate MSD for a scheduled UL and DL grant pair. For example, the UE may report MSD with respect to either real UE OTA REFSENS or a specified REFSENS, or even with respect to both. In some instances, the MSD may correspond to a REFSENS measured or determined by the UE and may correspond to a self-interference free baseline OTA REFSENS. In some instances, the MSD may correspond to a REFSENS measured or determined by the UE and may correspond to a specified REFSENS. The report of the MSD does not cause or increase the overhead in the network, such that a connected mode indication may be advantageous.
[0085] In some instances, a UE may indicate whether the MSD is caused by self-interference or by other or different interference, or by combination of the self-interference or by other or different interference. For example, the UE may provide an indication (e.g., MAC-CE) the portion of MSD that is cause by self-interference and the portion of the desensitization or degradation is due to the other or different interference. The other or different interference may be independent of resources that are allocated for communication between the network and the UE. In some aspects, the UE may indicate that the self-interference or the other or different interference is primary source of

MSD. In some aspects, the MSD may be reported as absolute dB values for each part (e.g., total MSD is 6 dB, where 3 dB from self-interference and 3 dB from other interference). In some instances, the MSD may be reported as ratio, or any other measure distinguishing the two MSD sources (e.g., total MSD is 6 dB, where a 50-50 ratio between the self-interference and the other or different interference is present). The UE may be configured to utilize implementation specific methods to improve the MSD over time, in which case it can indicate better MSD numbers over time.

[0086] FIG. **6** is a call flow diagram **600** of signaling between a UE **602** and a base station **604**. The base station **604** may be configured to provide at least one cell. The UE **602** may be configured to communicate with the base station **604**. For example, in the context of FIG. **1**, the base station **604** may correspond to base station **102** and the UE **602** may correspond to at least UE **104**. In another example, in the context of FIG. **3**, the base station **604** may correspond to base station **310** and the UE **602** may correspond to UE **350**.

[0087] At **606**, the base station **604** may provide a request for a measurement of an MSD. The base station **604** may provide the request for the measurement of the MSD to a UE **602**. The UE **602** may receive the request for the measurement of the MSD from the base station **604**.

[0088] At **608**, the UE **602** may measure the MSD. The UE may measure the MSD in response to receipt of the request for the measurement of the MSD from the base station **604**. The UE may measure the MSD caused by self-interference or a different interference than the self-interference. In some instances, the different interference other than the self-interference detected by the UE may include interference due to blockage of antenna arrays at the UE due to the user, such as but not limited to the user hand, head, or other body part blocking the antenna arrays at the UE.

[0089] At **610**, the UE **602** may determine an interference source associated with the MSD. The interference source indication reported for the MSD may be based on a determined interference source. For example, the interference source indication may indicate that the interference source may include the self-interference, the different interference other than the self-interference, or the combination of the self-interference and the different interference.

[0090] At **612**, the UE **602** may report a report comprising the measurement of the MSD. The UE **602** may provide the report comprising the measurement of the MSD to the base station **604**. The base station **604** may obtain the report comprising the measurement of the MSD from the UE **602**. The report comprising the measurement of the MSD may include an interference source indication of the MSD. The interference source indication may indicate that the MSD is due to the self-interference, the different interference, or a combination of the self-interference and the different interference. In some aspects, the interference source indication may indicate a primary interference source associated with the MSD. For example, the primary interference source associated with the MSD may be indicated as the self-interference, the different interference, or the combination of the self-interference and the different interference. In some aspects, the different interference is at least based on blockage of antenna elements of the UE.

[0091] In some aspects, the UE may report an amount of the MSD caused by a respective interference source to the base station. The base station may obtain the report indicating the amount of the MSD caused by the respective interference source from the UE. The UE may indicate in the report the amount of MSD that may be caused by the self-interference, the different interference, or the combination of the self-interference and the different interference.

[0092] In some aspects, the UE may report a first amount of the MSD and a second amount of the MSD to the base station. The base station may obtain the report indicating the first amount of the MSD and the second amount of the MSD from the UE. The first amount of the MSD may correspond to the amount of the MSD caused by the self-interference. The second amount of the MSD may correspond to the amount of the MSD caused by the different interference.

[0093] In some aspects, the UE may provide a report comprising the measurement of the MSD, including relative information associated with the first amount of the MSD caused by the self-interference. In some examples, the self-interference may include the interference caused by the transmission of the UE as an interferer. For example, the first amount of the MSD caused by self-interference may include the MSD caused by the transmission of the UE.

[0094] In some aspects, the relative information may be conveyed as channel state quality information or channel quality indicator (CQI), providing relative details about the degradation of the channel when the UE's own transmission acts as an interferer. In some aspects, the relative information may be included in a CQI. Table 2 shows an example of a CQI table that includes the MSD information in accordance with various aspects of the present disclosure. As shown in Table 2, the CQI value may include a self-interference field (e.g., S**1**, S**2**, . . . , S**14**). In some examples, the self-interference field may include a one-bit indication (e.g., a binary indication) indicating a presence or an absence of the first amount of the MSD caused by the self-interference (e.g., by the UE's transmission) in the reported value. For example, the value of “1” in this one-bit indication may indicate the presence of the first amount of the MSD caused by the self-interference, while the value of “0” may indicate the absence of the first amount of the MSD caused by the self-interference.

[0095] In some examples, the self-interference field (e.g., S**1**, S**2**, . . . , S**14**) may include a multi-bit indication indicating the change in the CQI resulting from the first amount of the MSD caused by the self-interference, for example, when the UE's frequency band for the transmission is scheduled with downlink resources. For example, the change in the CQI due to the first amount of the MSD caused by the self-interference may be represented as a number of CQI variation steps. Each CQI variation step may correspond to a decibel change, such as a change of 1.5 dB.

TABLE-US-00002 TABLE 2 An example of an amended CQI Table including the MSD information Self- CQI Index Modulation Code rate × 1024 Efficiency interference 0 Out of range 1 QPSK 78 0.1523 S.sub.1 2 QPSK 120 0.2344 S.sub.2 3 QPSK 193 0.3770 S.sub.3 4 QPSK 306 0.6016 S.sub.4 5 QPSK 449 0.8770 S.sub.5 6 QPSK 602 1.1758 S.sub.6 7 16QAM 378 1.4766 S.sub.7 8 16QAM 490 1.9141 S.sub.8 9 16QAM 616 2.4063 S.sub.9 10 64QAM 466 2.7305 S.sub.10 11 64QAM 567 3.3223 S.sub.11 12 64QAM 666 3.9023 S.sub.12 13 64QAM 772 4.5234 S.sub.13 14 64QAM 873 5.1152 S.sub.14 15 64QAM 948 5.5547 S.sub.15

[0096] In some aspects, the UE may report a ratio of the MSD. The UE may report the ratio of the MSD caused by the different interference sources to the base station. The base station may obtain the report indicating the ratio of the MSD caused by the different interference sources from the UE. For example, the report may indicate a ratio of the MSD caused by the self-interference and the different interference. The report may indicate a ratio of the MSD caused by the self-interference and the combination of the self-interference and the different interference. The report may indicate a ratio of the MSD caused by the different interference and the combination of the self-interference and the different interference.

[0097] In some aspects, the UE may report, to the base station, the MSD for a scheduled uplink and downlink grant pair measured at the UE. The base station may obtain, from the UE, the report indicting the MSD for the scheduled uplink and downlink grant pair measured at the UE. In some aspects, the request for the measurement of the MSD may include a request for a measurement of the MSD for the scheduled uplink and downlink grant pair. In such instances, the UE may report the MSD for the scheduled uplink and downlink grant pair measured at the UE. The report may include the interference source indication that the MSD is due to the self-interference, the different interference, or the combination of the self-interference and the different interference. In some aspects, the different interference may be independent of the scheduled uplink and downlink grant pair.

[0098] At **614**, the base station **604** may schedule communication based on the MSD. The base station may schedule the communication with the UE based on the MSD measured at the UE. In some aspects, the base station, in scheduling the communication based on the MSD, may schedule the communication with different frequency resources. The base station may schedule the communication with different frequency resources if the self-interference is indicated as a primary source of the MSD or causes a threshold amount of the MSD. In some aspects, the base station, in scheduling the communication based on the MSD, may maintain the scheduled communication. The base station may maintain the scheduled communication if the self-interference is not indicated as the primary source of the MSD or does not cause the threshold amount of the MSD.

[0099] At **616**, the UE and base station may communicate with each other. The UE and base station may communicate with each other utilizing the scheduled communication resources based on the MSD.

[0100] FIG. **7** is a flowchart **700** of a method of wireless communication. The method may be performed by a UE (e.g., the UE **104**; the apparatus **904**). One or more of the illustrated operations may be omitted, transposed, or contemporaneous. The method may allow a UE to report MSD based on the source of interference.

[0101] At **702**, the UE may receive a request for a measurement of an MSD. For example, **702** may be performed by measurement component **198** of apparatus **904**. The UE may receive the request for the measurement of the MSD from a network entity. FIG. **6** illustrates an example of a UE receiving a request for the measurement of MSD from a base station, for example.

[0102] At **704**, the UE may measure the MSD. For example, **704** may be performed by measurement component **198** of apparatus **904**. The UE may measure the MSD in response to receipt of the request for the measurement of the MSD. The UE may measure the MSD caused by self-interference or a different interference than the self-interference. In some instances, the different interference other than the self-interference detected by the UE may include interference due to blockage of antenna arrays at the UE due to the user, such as but not limited to the user hand, head, or other body part blocking the antenna arrays at the UE. FIG. **6** illustrates an example of a UE measuring the MSD, for example.

[0103] In some aspects, the UE may report a report comprising the measurement of the MSD. The UE may provide the report comprising the measurement of the MSD to the network entity. The report comprising the measurement of the MSD may include an interference source indication of the MSD. The interference source indication may indicate that the MSD is due to the self-interference, the different interference, or a combination of the self-interference and the different interference. In some aspects, the interference source indication may indicate a primary interference source associated with the MSD. For example, the primary interference source associated with the MSD may be indicated as the self-interference, the different interference, or the combination of the self-interference and the different interference. In some aspects, the different interference is at least based on blockage of antenna elements of the UE. FIG. **6** illustrates an example of a UE reporting the MSD to a base station.

[0104] In some aspects, at **706**, the UE may report a report comprising the measurement of the MSD including relative information associated with a first amount of the MSD caused by the self-interference. For example, **706** may be performed by measurement component **198** of apparatus **904**. In some examples, the self-interference may include the interference caused by the transmission of the UE as an interferer. For example, the first amount of the MSD caused by self-interference may include the MSD caused by the transmission of the UE.

[0105] FIG. **8** is a flowchart **800** of a method of wireless communication. The method may be performed by a UE (e.g., the UE **104**; the apparatus **904**). One or more of the illustrated operations may be omitted, transposed, or contemporaneous. The method may allow a UE to report MSD based on the source of interference.

[0106] At **802**, the UE may receive a request for a measurement of an MSD. For example, **802** may be performed by measurement component **198** of apparatus **904**. The UE may receive the request for the measurement of the MSD from a network entity.

[0107] At **804**, the UE may measure the MSD. For example, **804** may be performed by measurement component **198** of apparatus **904**. The UE may measure the MSD in response to receipt of the request for the measurement of the MSD. The UE may measure the MSD caused by self-interference or a different interference than the self-interference. In some instances, the different interference other than the self-interference detected by the UE may include interference due to blockage of antenna arrays at the UE due to the user, such as but not limited to the user hand, head, or other body part blocking the antenna arrays at the UE.

[0108] At **806**, the UE may determine an interference source associated with the MSD. For example, **806** may be performed by measurement component **198** of apparatus **904**. The interference source indication reported for the MSD may be based on a determined interference source. For example, the interference source indication may indicate that the interference source may include the self-interference, the different interference other than the self-interference, or the combination of the self-interference and the different interference.

[0109] At **808**, the UE may report a report comprising the measurement of the MSD. For example, **808** may be performed by measurement component **198** of apparatus **904**. The UE may provide the report comprising the measurement of the MSD to the network entity. The report comprising the measurement of the MSD may include an interference source indication of the MSD. The interference source indication may indicate that the MSD is due to the self-interference, the different interference, or a combination of the self-interference and the different interference. In some aspects, the interference source indication may indicate a primary interference source associated with the MSD. For example, the primary interference source associated with the MSD may be indicated as the self-interference, the different interference, or the combination of the self-interference and the different interference. In some aspects, the different interference is at least based on blockage of antenna elements of the UE.

[0110] In some aspects, the UE may report a report comprising the measurement of the MSD including relative information associated with a first amount of the MSD caused by the self-interference.

[0111] In some examples, the self-interference may include the interference caused by the transmission of the UE as an interferer. For example, the first amount of the MSD caused by self-interference may include the MSD caused by the transmission of the UE.

[0112] In some aspects, the relative information may be included in the CQI. For example, as shown in Table 2, the CQI value may include a self-interference field (e.g., S**1**, S**2**, . . . , S**14**). In some examples, the self-interference field may include a one-bit indication (e.g., a binary indication) indicating a presence or an absence of the first amount of the MSD caused by the self-interference (e.g., by the UE's transmission) in the reported value. For example, the value of “1” in this one-bit indication may indicate the presence of the first amount of the MSD caused by the self-interference, while the value of “0” may indicate the absence of the first amount of the MSD caused by the self-interference.

[0113] In some examples, the self-interference field (e.g., S**1**, S**2**, . . . , S**14**) may include a multi-bit indication indicating the change in the CQI resulting from the first amount of the MSD caused by the self-interference, for example, when the UE's frequency band for the transmission is scheduled with downlink resources. For example, the change in the CQI due to the first amount of the MSD caused by the self-interference may be represented as a number of CQI variation steps. For example, each CQI variation step may correspond to a decibel change, such as a change of 1.5 dB.

[0114] At **810**, the UE may report an amount of the MSD caused by a respective interference source. For example, **810** may be performed by measurement component **198** of apparatus **904**. The UE may indicate in the report the amount of MSD that may be caused by the self-interference, the different interference, or the combination of the self-interference and the different interference.

[0115] At **812**, the UE may report a first amount of the MSD and a second amount of the MSD. For example, **812** may be performed by measurement component **198** of apparatus **904**. The first amount of the MSD may correspond to the amount of the MSD caused by the self-interference. The second amount of the MSD may correspond to the amount of the MSD caused by the different interference.

[0116] At **814**, the UE may report a ratio of the MSD. For example, **814** may be performed by measurement component **198** of apparatus **904**. The UE may report the ratio of the MSD caused by the different interference sources. For example, the report may indicate a ratio of the MSD caused by the self-interference and the different interference. The report may indicate a ratio of the MSD caused by the self-interference and the combination of the self-interference and the different interference. The report may indicate a ratio of the MSD caused by the different interference and the combination of the self-interference and the different interference.

[0117] At **816**, the UE may report the MSD for a scheduled uplink and downlink grant pair measured at the UE. For example, **816** may be performed by measurement component **198** of apparatus **904**. In some aspects, the request for the measurement of the MSD may include a request for a measurement of the MSD for the scheduled uplink and downlink grant pair. In such instances, the UE may report the MSD for the scheduled uplink and downlink grant pair measured at the UE. The report may include the interference source indication that the MSD is due to the self-interference, the different interference, or the combination of the self-interference and the different interference. In some aspects, the different interference may be independent of the scheduled uplink and downlink grant pair.

[0118] FIG. **9** is a diagram **900** illustrating an example of a hardware implementation for an apparatus **904**. The apparatus **904** may be a UE, a component of a UE, or may implement UE functionality. In some aspects, the apparatus **904** may include at least one cellular baseband processor **924** (also referred to as a modem) coupled to one or more transceivers **922** (e.g., cellular RF transceiver). The cellular baseband processor(s) **924** may include at least one on-chip memory **924′**. In some aspects, the apparatus **904** may further include one or more subscriber identity modules (SIM) cards **920** and at least one application processor **906** coupled to a secure digital (SD) card **908** and a screen **910**. The application processor(s) **906** may include on-chip memory **906′**. In some aspects, the apparatus **904** may further include a Bluetooth module **912**, a WLAN module **914**, an SPS module **916** (e.g., GNSS module), one or more sensor modules **918** (e.g., barometric pressure sensor/altimeter; motion sensor such as inertial measurement unit (IMU), gyroscope, and/or accelerometer(s); light detection and ranging (LIDAR), radio assisted detection and ranging (RADAR), sound navigation and ranging (SONAR), magnetometer, audio and/or other technologies used for positioning), additional memory modules **926**, a power supply **930**, and/or a camera **932**. The Bluetooth module **912**, the WLAN module **914**, and the SPS module **916** may include an on-chip transceiver (TRX) (or in some cases, just a receiver (RX)). The Bluetooth module **912**, the WLAN module **914**, and the SPS module **916** may include their own dedicated antennas and/or utilize the antennas **980** for communication. The cellular baseband processor(s) **924** communicates through the transceiver(s) **922** via one or more antennas **980** with the UE **104** and/or with an RU associated with a network entity **902**. The cellular baseband processor(s) **924** and the application processor(s) **906** may each include a computer-readable medium/memory **924′**, **906′**, respectively. The additional memory modules **926** may also be considered a computer-readable medium/memory. Each computer-readable medium/memory **924′**, **906′**, memory modules **926** may be non-transitory. The cellular baseband processor(s) **924** and the application processor(s) **906** are each responsible for general processing, including the execution of software stored on the computer-readable medium/memory. The software, when executed by the cellular baseband processor(s) **924**/application processor(s) **906**, causes the cellular baseband processor(s) **924**/application processor(s) **906** to perform the various functions described supra. The cellular baseband processor(s) **924** and the application processor(s) **1006** are configured to perform the various functions described supra based at least in part of the information stored in the memory. That is, the cellular baseband processor(s) **924** and the application processor(s) **906** may be configured to perform a first subset of the various functions described supra without information stored in the memory and may be configured to perform a second subset of the various functions described supra based on the information stored in the memory. The computer-readable medium/memory may also be used for storing data that is manipulated by the cellular baseband processor(s) **924**/application processor(s) **906** when executing software. The cellular baseband processor(s) **924**/application processor(s) **906** may be a component of the UE **350** and may include the at least one memory **360** and/or at least one of the TX processor **368**, the RX processor **356**, and the controller/processor **359**. In one configuration, the apparatus **904** may be at least one processor chip (modem and/or application) and include just the cellular baseband processor(s) **924** and/or the application processor(s) **906**, and in another configuration, the apparatus **904** may be the entire UE (e.g., see UE **350** of FIG. **3**) and include the additional modules of the apparatus **904**.

[0119] As discussed supra, in some aspects, the component **198** may be configured to receive a request for a measurement of an MSD; measure the MSD caused by self-interference or a different interference than the self-interference; and report a report comprising the measurement of the MSD including an interference source indication that the MSD is due to the self-interference, the different interference, or a combination of the self-interference and the different interference. In some aspects, the component **198** may be configured to receive a request for a measurement of an MSD; measure the MSD caused by self-interference or a different interference than the self-interference; and report a report comprising the measurement of the MSD including relative information associated with a first amount of the MSD caused by the self-interference. The component **198** and/or the apparatus **904** may be further configured to perform any of the aspects described in connection with the flowcharts in FIGS. **7** and/or **8** and/or any of the aspects performed by the UE in the communication flow in FIG. **6**. The component **198** may be within the cellular baseband processor(s) **924**, the application processor(s) **906**, or both the cellular baseband processor(s) **924** and the application processor(s) **906**. The component **198** may be one or more hardware components specifically configured to carry out the stated processes/algorithm, implemented by one or more processors configured to perform the stated processes/algorithm, stored within a computer-readable medium for implementation by one or more processors, or some combination thereof. When multiple processors are implemented, the multiple processors may perform the stated processes/algorithm individually or in combination. As shown, the apparatus **904** may include a variety of components configured for various functions. In one configuration, the apparatus **904**, and in particular the cellular baseband processor(s) **924** and/or the application processor(s) **906**, may include means for receiving a request for a measurement of an MSD. The apparatus includes means for measuring the MSD caused by self-interference or a different interference than the self-interference. The apparatus includes means for reporting a report comprising the measurement of the MSD including an interference source indication that the MSD is due to the self-interference, the different interference, or a combination of the self-interference and the different interference. The apparatus includes means for reporting a report comprising the measurement of the MSD including relative information associated with a first amount of the MSD caused by the self-interference. The apparatus further includes means for determining an interference source associated with the MSD, wherein the interference source indication reported for the MSD is based on a determined interference source. The apparatus further includes means for reporting an amount of the MSD caused by a respective interference source. The apparatus further includes means for reporting a first amount of the MSD caused by the self-interference and a second amount of the MSD caused by the different interference. The apparatus further includes means for reporting a ratio of the MSD caused by different interference sources. The apparatus further includes means for reporting the MSD for the scheduled uplink and downlink grant pair measured at the UE, the report including the interference source indication that the MSD is due to the self-interference, the different interference, or the combination of the self-interference and the different interference. The apparatus **904** may include means for performing any of the aspects described in connection with the flowcharts in FIGS. **7** and/or **8** and/or any of the aspects performed by the UE in the communication flow in FIG. **6**. The means may be the component **198** of the apparatus **904** configured to perform the functions recited by the means. As described supra, the apparatus **904** may include the TX processor **368**, the RX processor **356**, and the controller/processor **359**. As such, in one configuration, the means may be the TX processor **368**, the RX processor **356**, and/or the controller/processor **359** configured to perform the functions recited by the means.

[0120] FIG. **10** is a flowchart **1000** of a method of wireless communication. The method may be performed by a network node or network entity such as a base station or one or more components of a base station (e.g., the base station **102**, **310**, **604**; the CU **110**; the DU **130**; the RU **140**; the network entity **1202**). One or more of the illustrated operations may be omitted, transposed, or contemporaneous. The method may enable the network to configure a UE to report MSD based on the source of interference.

[0121] At **1002**, the network entity may provide a request for a measurement of an MSD. For example, **1002** may be performed by measurement component **199** of network entity **1202**. The network entity may provide the request for the measurement of the MSD to a UE.

[0122] In some aspects, the network entity may obtain a report including the measurement of the MSD. The report may include the measurement of the MSD caused by self-interference at the UE or a different interference than the self-interference at the UE. The report may include an interference source indication that the MSD is due to the self-interference, the different interference, or a combination of the self-interference and the different interference. In some aspects, the report of the MSD may include a determination of an interference source associated with the MSD. The interference source indication reported for the MSD may be based on a determined interference source. For example, the interference source indication may indicate that the MSD is due to the self-interference, the different interference, or a combination of the self-interference and the different interference. In some aspects, the interference source indication may indicate a primary interference source associated with the MSD. For example, the primary interference source associated with the MSD may be indicated as the self-interference, the different interference, or the combination of the self-interference and the different interference. In some aspects, the different interference is at least based on blockage of antenna elements of the UE.

[0123] In some aspects, at **1004**, the network entity may obtain a report including the measurement of the MSD caused by self-inference at the UE or a different interference than the self-interference at the UE. The report may include relative information associated with a first amount of the MSD caused by the self-interference. For example, **1004** may be performed by measurement component **199** of network entity **1202**. In some examples, the self-interference may include the interference caused by a transmission of the UE. For example, the first amount of the MSD caused by self-interference may include the MSD caused by the transmission of the UE.

[0124] At **1006**, the network entity may schedule communication based on the MSD. For example, **1006** may be performed by measurement component **199** of network entity **1202**. The network entity may schedule the communication with the UE based on the MSD measured at the UE.

[0125] FIG. **11** is a flowchart **1100** of a method of wireless communication. The method may be performed by a network node or network entity such as a base station or one or more components of a base station (e.g., the base station **102**, **310**, **604**; the CU **110**; the DU **130**; the RU **140**; the network entity **1202**,). One or more of the illustrated operations may be omitted, transposed, or contemporaneous. The method may enable the network to configure a UE to report MSD based on the source of interference.

[0126] At **1102**, the network entity may provide a request for a measurement of an MSD. For example, **1102** may be performed by measurement component **199** of network entity **1202**. The network entity may provide the request for the measurement of the MSD to a UE. FIG. **6** illustrates an example of a base station **604** transmitting a request to a UE **602**.

[0127] At **1104**, the network entity may obtain a report including the measurement of the MSD. For example, **1104** may be performed by measurement component **199** of network entity **1202**. FIG. **6**

illustrates an example of a base station receiving a report, at **612**, for example. The report may include the measurement of the MSD caused by self-interference at the UE or a different interference than the self-interference at the UE. The report may include an interference source indication that the MSD is due to the self-interference, the different interference, or a combination of the self-interference and the different interference. In some aspects, the report of the MSD may include a determination of an interference source associated with the MSD. The interference source indication reported for the MSD may be based on a determined interference source. For example, the interference source indication may indicate that the MSD is due to the self-interference, the different interference, or a combination of the self-interference and the different interference. In some aspects, the interference source indication may indicate a primary interference source associated with the MSD. For example, the primary interference source associated with the MSD may be indicated as the self-interference, the different interference, or the combination of the self-interference and the different interference. In some aspects, the different interference is at least based on blockage of antenna elements of the UE. In some aspects, the report may include relative information associated with a first amount of the MSD caused by the self-interference. In some examples, the self-interference may include the interference caused by a transmission of the UE.
[0128] In some aspects, the relative information may be included in the CQI. For example, as shown in Table 2, the CQI value may include a self-interference field (e.g., S**1**, S**2**, . . . , S**14**). In some examples, the self-interference field may include a one-bit indication (e.g., a binary indication) indicating the presence or the absence of the first amount of the MSD caused by the self-interference (e.g., by the UE's transmission) in the reported value. For example, the value of “1” in this one-bit indication may indicate the presence of the first amount of the MSD caused by the self-interference, while the value of “0” may indicate the absence of the first amount of the MSD caused by the self-interference.
[0129] In some examples, the self-interference field (e.g., S**1**, S**2**, . . . , S**14**) may include a multi-bit indication indicating the change in the CQI resulting from the first amount of the MSD caused by the self-interference, for example, when the UE's frequency band for the transmission is scheduled with downlink resources. For example, the change in the CQI due to the first amount of the MSD caused by the self-interference may be represented as a number of CQI variation steps. For example, each CQI variation step may correspond to a decibel change, such as a change of 1.5 dB.
[0130] At **1106**, the network entity may obtain the report indicating an amount of the MSD caused by a respective interference source. For example, **1106** may be performed by measurement component **199** of network entity **1202**. The report may indicate in the report the amount of MSD that may be caused by the self-interference, the different interference, or the combination of the self-interference and the different interference. In some aspects, the report may include a first amount of the MSD caused by the self-interference and may include a second amount of the MSD caused by the different interference.
[0131] At **1108**, the network entity may obtain the report indicating a ratio of the MSD caused by different interference sources. For example, **1108** may be performed by measurement component **199** of network entity **1202**. The report may indicate the ratio of the MSD caused by the different interference sources. For example, the report may indicate a ratio of the MSD caused by the self-interference and the different interference. The report may indicate a ratio of the MSD caused by the self-interference and the combination of the self-interference and the different interference. The report may indicate a ratio of the MSD caused by the different interference and the combination of the self-interference and the different interference.
[0132] At **1110**, the network entity may obtain the MSD for a scheduled uplink and downlink grant pair measured at the UE. For example, **1110** may be performed by measurement component **199** of network entity **1202**. In some aspects, the request for the measurement of the MSD may include a request for a measurement of the MSD for the scheduled uplink and downlink grant pair. In such instances, the report may indicate the MSD for the scheduled uplink and downlink grant pair

measured at the UE. The report may include the interference source indication that the MSD is due to the self-interference, the different interference, or the combination of the self-interference and the different interference. In some aspects, the different interference may be independent of the scheduled uplink and downlink grant pair.

[0133] At **1112**, the network entity may schedule communication based on the MSD. For example, **1112** may be performed by measurement component **199** of network entity **1202**. The network entity may schedule the communication with the UE based on the MSD measured at the UE.

[0134] At **1114**, the network entity, in scheduling the communication based on the MSD, may schedule the communication with different frequency resources. For example, **1114** may be performed by measurement component **199** of network entity **1202**. The network entity may schedule the communication with different frequency resources if the self-interference is indicated as a primary source of the MSD or causes a threshold amount of the MSD.

[0135] At **1116**, the network entity, in scheduling the communication based on the MSD, may maintain the scheduled communication. For example, **1116** may be performed by measurement component **199** of network entity **1202**. The network entity may maintain the scheduled communication if the self-interference is not indicated as the primary source of the MSD or does not cause the threshold amount of the MSD.

[0136] FIG. **12** is a diagram **1200** illustrating an example of a hardware implementation for a network entity **1202**. The network entity **1202** may be a BS, a component of a BS, or may implement BS functionality. The network entity **1202** may include at least one of a CU **1210**, a DU **1230**, or an RU **1240**. For example, depending on the layer functionality handled by the component **199**, the network entity **1202** may include the CU **1210**; both the CU **1210** and the DU **1230**; each of the CU **1210**, the DU **1230**, and the RU **1240**; the DU **1230**; both the DU **1230** and the RU **1240**; or the RU **1240**. The CU **1210** may include at least one CU processor **1212**. The CU processor(s) **1212** may include on-chip memory **1212′**. In some aspects, the CU **1210** may further include additional memory modules **1214** and a communications interface **1218**. The CU **1210** communicates with the DU **1230** through a midhaul link, such as an F1 interface. The DU **1230** may include at least one DU processor **1232**. The DU processor(s) **1232** may include on-chip memory **1232′**. In some aspects, the DU **1230** may further include additional memory modules **1234** and a communications interface **1238**. The DU **1230** communicates with the RU **1240** through a fronthaul link. The RU **1240** may include at least one RU processor **1242**. The RU processor(s) **1242** may include on-chip memory **1242′**. In some aspects, the RU **1240** may further include additional memory modules **1244**, one or more transceivers **1246**, antennas **1280**, and a communications interface **1248**. The RU **1240** communicates with the UE **104**. The on-chip memory **1212′**, **1232′**, **1242′** and the additional memory modules **1214**, **1234**, **1244** may each be considered a computer-readable medium/memory. Each computer-readable medium/memory may be non-transitory. Each of the processors **1212**, **1232**, **1242** is responsible for general processing, including the execution of software stored on the computer-readable medium/memory. The software, when executed by the corresponding processor(s) causes the processor(s) to perform the various functions described supra. The computer-readable medium/memory may also be used for storing data that is manipulated by the processor(s) when executing software.

[0137] As discussed supra, in some aspects, the component **199** may be configured to provide a request for a measurement of an MSD; obtain a report comprising the measurement of the MSD caused by self-interference at a UE or a different interference than the self-interference at the UE, wherein the report includes an interference source indication that the MSD is due to the self-interference, the different interference, or a combination of the self-interference and the different interference; and schedule communication based on the MSD. In some aspects, the component **199** may be configured to provide a request for a measurement of an MSD; obtain a report comprising the measurement of the MSD caused by self-interference at a UE or a different interference than the self-interference at the UE, wherein the report includes relative information associated with a first amount of the MSD caused by the self-interference; and schedule communication based on the MSD. The component **199** and/or the network entity may be further configured to perform any of the aspects described in connection with the flowcharts in FIGS. **10** and/or **11** and/or any of the aspects performed by the base station in the communication flow in FIG. **6**. The component **199** may be within one or more processors of one or more of the CU **1210**, DU **1230**, and the RU **1240**. The component **199** may be one or more hardware components specifically configured to carry out the stated processes/algorithm, implemented by one or more processors configured to perform the stated processes/algorithm, stored within a computer-readable medium for implementation by one or more processors, or some combination thereof. When multiple processors are implemented, the multiple processors may perform the stated processes/algorithm individually or in combination. The network entity **1202** may include a variety of components configured for various functions. In one configuration, the network entity **1202** may include means for providing a request for a measurement of an MSD. The network entity includes means for obtaining a report comprising the measurement of the MSD caused by self-interference at a UE or a different interference than the self-interference at the UE, wherein the report includes an interference source indication that the MSD is due to the self-interference, the different interference, or a combination of the self-interference and the different interference. The network entity includes means for obtaining a report comprising the measurement of the MSD caused by self-interference at a UE or a different interference than the self-interference at the UE, wherein the report includes relative information associated with a first amount of the MSD caused by the self-interference. The network entity includes means for scheduling communication based on the MSD. The network entity further includes means for obtaining the report indicating an amount of the MSD caused by a respective interference source. The network entity further includes means for obtaining the report indicating a ratio of the MSD caused by different interference sources. The network entity further includes means for scheduling the communication with different frequency resources if the self-interference is indicated as a primary source of the MSD or causes a threshold amount of the MSD. The network entity further includes means for maintaining the scheduled communication (e.g., maintaining previously scheduled communication, if the self-interference is not indicated as the primary source of the MSD or does not cause the threshold amount of the MSD. The network entity further includes means for obtaining the MSD for the scheduled uplink and downlink grant pair measured at the UE, the report including the interference source indication that the MSD is due to the self-interference, the different interference, or the combination of the self-interference and the different interference. The network entity may further include means for performing any of the aspects described in connection with the flowcharts in FIGS. **10** and/or **11** and/or any of the aspects performed by the base station in the communication flow in FIG. **6**. The means may be the component **199** of the network entity **1202** configured to perform the functions recited by the means. As described supra, the network entity **1202** may include the TX processor **316**, the RX processor **370**, and the controller/processor **375**. As such, in one configuration, the means may be the TX processor **316**, the RX processor **370**, and/or the controller/processor **375** configured to perform the functions recited by the means.

[0138] It is understood that the specific order or hierarchy of blocks in the processes/flowcharts disclosed is an illustration of example approaches. Based upon design preferences, it is understood that the specific order or hierarchy of blocks in the processes/flowcharts may be rearranged. Further, some blocks may be combined or omitted. The accompanying method claims present elements of the various blocks in a sample order, and are not limited to the specific order or hierarchy presented.

[0139] The previous description is provided to enable any person skilled in the art to practice the various aspects described herein. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects. Thus, the claims are not limited to the aspects described herein, but are to be accorded the full scope consistent with the language claims. Reference to an element in the singular does not mean "one and only one" unless specifically so stated, but rather "one or more." Terms such as "if," "when," and "while" do not imply an immediate temporal relationship or reaction. That is, these phrases, e.g., "when," do not imply an immediate action in response to or during the occurrence of an action, but simply imply that if a condition is met then an action will occur, but without requiring a specific or immediate time constraint for the action to occur. The word "exemplary" is used herein to mean "serving as an example, instance, or illustration." Any aspect described herein as "exemplary" is not necessarily to be construed as preferred or advantageous over other aspects. Unless specifically stated otherwise, the term "some" refers to one or more. Combinations such as "at least one of A, B, or C," "one or more of A, B, or C," "at least one of A, B, and C," "one or more of A, B, and C," and "A, B, C, or any combination thereof" include any combination of A, B, and/or C, and may include multiples of A, multiples of B, or multiples of C. Specifically, combinations such as "at least one of A, B, or C," "one or more of A, B, or C," "at least one of A, B, and C," "one or more of A, B, and C," and "A, B, C, or any combination thereof" may be A only, B only, C only, A and B, A and C, B and C, or A and B and C, where any such combinations may contain one or more member or members of A, B, or C. Sets should be interpreted as a set of elements where the elements number one or more. Accordingly, for a set of X, X would include one or more elements. When at least one processor is configured to perform a set of functions, the at least one processor, individually or in any combination, is configured to perform the set of functions. Accordingly, each processor of the at least one processor may be configured to perform a particular subset of the set of functions, where the subset is the full set, a proper subset of the set, or an empty subset of the set. If a first apparatus receives data from or transmits data to a second apparatus, the data may be received/transmitted directly between the first and second apparatuses, or indirectly between the first and second apparatuses through a set of apparatuses. A device configured to "output" data, such as a transmission, signal, or message, may transmit the data, for example with a transceiver, or may send the data to a device that transmits the data. A device configured to "obtain" data, such as a transmission, signal, or message, may receive, for example with a transceiver, or may obtain the data from a device that receives the data. Information stored in a memory includes instructions and/or data. All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are encompassed by the claims. Moreover, nothing disclosed herein is dedicated to the public regardless of whether such disclosure is explicitly recited in the claims. The words "module," "mechanism," "element," "device," and the like may not be a substitute for the word "means." As such, no claim element is to be construed as a means plus function unless the element is expressly recited using the phrase "means for."

[0140] As used herein, the phrase "based on" shall not be construed as a reference to a closed set of information, one or more conditions, one or more factors, or the like. In other words, the phrase "based on A" (where "A" may be information, a condition, a factor, or the like) shall be construed as "based at least on A" unless specifically recited differently.

[0141] The following aspects are illustrative only and may be combined with other aspects or teachings described herein, without limitation.

[0142] Aspect 1 is a method of wireless communication at a UE comprising receiving a request for a measurement of an MSD; measuring the MSD caused by self-interference or a different interference than the self-interference; and reporting a report comprising the measurement of the MSD including an interference source indication that the MSD is due to the self-interference, the different interference, or a combination of the self-interference and the different interference.

[0143] Aspect 2 is the method of aspect 1, further including determining an interference source associated with the MSD, wherein the interference source indication reported for the MSD is based on a determined interference source.

[0144] Aspect 3 is the method of any of aspects 1 and 2, further includes that the interference source indication indicates a primary interference source associated with the MSD.
[0145] Aspect 4 is the method of any of aspects 1-3, further including reporting an amount of the MSD caused by a respective interference source.
[0146] Aspect 5 is the method of any of aspects 1-4, further including reporting a first amount of the MSD caused by the self-interference and a second amount of the MSD caused by the different interference.
[0147] Aspect 6 is the method of any of aspects 1-5, further including reporting a ratio of the MSD caused by different interference sources.
[0148] Aspect 7 is the method of any of aspects 1-6, further includes that the different interference is at least based on blockage of antenna elements of the UE.
[0149] Aspect 8 is the method of any of aspects 1-7, further includes that the request is for the measurement of the MSD for a scheduled uplink and downlink grant pair, further including reporting the MSD for the scheduled uplink and downlink grant pair measured at the UE, the report including the interference source indication that the MSD is due to the self-interference, the different interference, or the combination of the self-interference and the different interference.
[0150] Aspect 9 is the method of any of aspects 1-8, further includes that the different interference is independent of the scheduled uplink and downlink grant pair.
[0151] Aspect 10 is an apparatus for wireless communication at a first wireless device including at least one processor coupled to a memory and at least one transceiver, the at least one processor configured to implement any of aspects 1-9.
[0152] Aspect 11 is an apparatus for wireless communication at a first wireless device including means for implementing any of aspects 1-9.
[0153] Aspect 12 is a computer-readable medium storing computer executable code, where the code when executed by a processor causes the processor to implement any of aspects 1-9.
[0154] Aspect 13 is a method for wireless communication at a network entity comprising providing a request for a measurement of an MSD; obtaining a report comprising the measurement of the MSD caused by self-interference at a UE or a different interference than the self-interference at the UE, wherein the report includes an interference source indication that the MSD is due to the self-interference, the different interference, or a combination of the self-interference and the different interference; and scheduling communication based on the MSD.
[0155] Aspect 14 is the method of aspect 13, further includes that the report of the MSD includes a determination of an interference source associated with the MSD, wherein the interference source indication reported for the MSD is based on a determined interference source.
[0156] Aspect 15 is the method of any of aspects 13 and 14, further includes that the interference source indication indicates a primary interference source associated with the MSD.
[0157] Aspect 16 is the method of any of aspects 13-15, further including obtaining the report indicating an amount of the MSD caused by a respective interference source.
[0158] Aspect 17 is the method of any of aspects 13-16, further includes that the report includes a first amount of the MSD caused by the self-interference and a second amount of the MSD caused by the different interference.
[0159] Aspect 18 is the method of any of aspects 13-17, further including obtaining the report indicating a ratio of the MSD caused by different interference sources.
[0160] Aspect 19 is the method of any of aspects 13-18, further includes that the different interference is at least based on blockage of antenna elements of the UE.
[0161] Aspect 20 is the method of any of aspects 13-19, further includes that to schedule the communication based on the MSD further including scheduling the communication with different frequency resources if the self-interference is indicated as a primary source of the MSD or causes a threshold amount of the MSD, or maintaining previously scheduled communication if the self-interference is not indicated as the primary source of the MSD or does not cause the threshold

amount of the MSD.
[0162] Aspect 21 is the method of any of aspects 13-20, further includes that the request is for the measurement of the MSD for a scheduled uplink and downlink grant pair further including obtaining the MSD for the scheduled uplink and downlink grant pair measured at the UE, the report including the interference source indication that the MSD is due to the self-interference, the different interference, or the combination of the self-interference and the different interference.
[0163] Aspect 22 is the method of any of aspects 13-21, further includes that the different interference is independent of the scheduled uplink and downlink grant pair.
[0164] Aspect 23 is an apparatus for wireless communication at a network entity including at least one processor coupled to a memory and at least one transceiver, the at least one processor configured to implement any of aspects 13-22.
[0165] Aspect 24 is an apparatus for wireless communication at a network entity including means for implementing any of aspects 13-22.
[0166] Aspect 25 is a computer-readable medium storing computer executable code, where the code when executed by a processor causes the processor to implement any of aspects 13-22.
[0167] Aspect 26 is a method of wireless communication at a UE comprising receiving a request for a measurement of an MSD; measuring the MSD caused by self-interference or a different interference than the self-interference; and reporting a report comprising the measurement of the MSD including relative information associated with a first amount of the MSD caused by the self-interference.
[0168] Aspect 27 is the method of aspect 26, wherein the self-interference includes an interference caused by a transmission of the UE.
[0169] Aspect 28 is the method of aspect 27, wherein the relative information is comprised in a channel quality indicator (CQI).
[0170] Aspect 29 is the method of aspect 28, wherein the CQI includes a binary indication indicating a presence or an absence of the first amount of the MSD caused by the self-interference in the measurement of the MSD.
[0171] Aspect 30 is the method of aspect 28, wherein the CQI includes a multi-bit indication indicating a change in the CQI due to the first amount of the MSD caused by the self-interference.
[0172] Aspect 31 is the method of aspect 30, wherein a frequency band of the transmission of the UE is scheduled with downlink resources.
[0173] Aspect 32 is the method of aspect 30, wherein the change in the CQI due to the first amount of the MSD caused by the self-interference includes a number of CQI variation steps.
[0174] Aspect 33 is the method of aspect 32, wherein each CQI variation step in the number of CQI variation steps represents a decibel change.
[0175] Aspect 34 is the method of aspect 26, further comprising reporting a ratio of the MSD caused by different interference sources.
[0176] Aspect 35 is an apparatus for wireless communication at a first wireless device including at least one processor coupled to a memory and at least one transceiver, the at least one processor configured to implement any of aspects 26-34.
[0177] Aspect 36 is an apparatus for wireless communication at a first wireless device including means for implementing any of aspects 26-34.
[0178] Aspect 37 is a computer-readable medium storing computer executable code, where the code when executed by a processor causes the processor to implement any of aspects 26-34.
[0179] Aspect 38 is a method for wireless communication at a network entity comprising providing a request for a measurement of an MSD; obtaining a report comprising the measurement of the MSD caused by self-interference at a UE or a different interference than the self-interference at the UE, wherein the report includes relative information associated with a first amount of the MSD caused by the self-interference; and scheduling communication based on the MSD.
[0180] Aspect 39 is the method of aspect 38, wherein the self-interference includes an interference

caused by a transmission of the UE.
[0181] Aspect 40 is the method of aspect 39, wherein the relative information is comprised in a channel quality indicator (CQI).
[0182] Aspect 41 is the method of aspect 40, wherein the CQI includes a binary indication indicating a presence or an absence of the first amount of the MSD caused by the self-interference in the measurement of the MSD.
[0183] Aspect 42 is the method of aspect 40, wherein the CQI includes a multi-bit indication indicating a change in the CQI due to the first amount of the MSD caused by the self-interference.
[0184] Aspect 43 is an apparatus for wireless communication at a network entity including at least one processor coupled to a memory and at least one transceiver, the at least one processor configured to implement any of aspects 38-42.
[0185] Aspect 44 is an apparatus for wireless communication at a network entity including means for implementing any of aspects 38-42.
[0186] Aspect 45 is a computer-readable medium storing computer executable code, where the code when executed by a processor causes the processor to implement any of aspects 38-42.

Claims

1. An apparatus for wireless communication at a user equipment (UE), comprising: at least one memory; and at least one processor coupled to the at least one memory and, based at least in part on information stored in the at least one memory, the at least one processor, individually or in any combination, is configured to cause the apparatus to: receive a request for a measurement of a maximum sensitivity degradation (MSD); measure the MSD caused by self-interference or a different interference than the self-interference; and report a report comprising the measurement of the MSD including relative information associated with a first amount of the MSD caused by the self-interference.
2. The apparatus of claim 1, further comprising a transceiver coupled to the at least one processor, the transceiver being configured to: receive the request for the measurement of the MSD; and report the report comprising the measurement of the MSD including the relative information associated with the first amount of the MSD caused by the self-interference.
3. The apparatus of claim 1, wherein the self-interference includes an interference caused by a transmission of the UE.
4. The apparatus of claim 3, wherein the relative information is comprised in a channel quality indicator (CQI).
5. The apparatus of claim 4, wherein the CQI includes a binary indication indicating a presence or an absence of the first amount of the MSD caused by the self-interference in the measurement of the MSD.
6. The apparatus of claim 4, wherein the CQI includes a multi-bit indication indicating a change in the CQI due to the first amount of the MSD caused by the self-interference.
7. The apparatus of claim 6, wherein a frequency band of the transmission of the UE is scheduled with downlink resources.
8. The apparatus of claim 6, wherein the change in the CQI due to the first amount of the MSD caused by the self-interference includes a number of CQI variation steps.
9. The apparatus of claim 8, wherein each CQI variation step in the number of CQI variation steps represents a decibel change.
10. The apparatus of claim 1, wherein the at least one processor is configured to: report a ratio of the MSD caused by different interference sources.
11. The apparatus of claim 1, wherein the different interference is at least based on blockage of antenna elements of the UE.
12. A method of wireless communication at a user equipment (UE), comprising: receiving a request

for a measurement of a maximum sensitivity degradation (MSD); measuring the MSD caused by self-interference or a different interference than the self-interference; and reporting a report comprising the measurement of the MSD including relative information associated with a first amount of the MSD caused by the self-interference.

13. The method of claim 12, wherein the self-interference includes an interference caused by a transmission of the UE.

14. The method of claim 13, wherein the relative information is comprised in a channel quality indicator (CQI).

15. The method of claim 14, wherein the CQI includes a binary indication indicating a presence or an absence of the first amount of the MSD caused by the self-interference in the measurement of the MSD.

16. The method of claim 14, wherein the CQI includes a multi-bit indication indicating a change in the CQI due to the first amount of the MSD caused by the self-interference.

17. The method of claim 16, wherein a frequency band of the transmission of the UE is scheduled with downlink resources.

18. The method of claim 16, wherein the change in the CQI due to the first amount of the MSD caused by the self-interference includes a number of CQI variation steps.

19. The method of claim 18, wherein each CQI variation step in the number of CQI variation steps represents a decibel change.

20. An apparatus for wireless communication at a network entity, comprising: at least one memory; and at least one processor coupled to the at least one memory and, based at least in part on information stored in the at least one memory, the at least one processor, individually or in any combination, is configured to cause the apparatus to: provide a request for a measurement of a maximum sensitivity degradation (MSD); obtain a report comprising the measurement of the MSD caused by self-interference at a user equipment (UE) or a different interference than the self-interference at the UE, wherein the report includes relative information associated with a first amount of the MSD caused by the self-interference; and schedule communication based on the MSD.